



College of Business

Assignment # 1

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1. Introduction

Business Intelligence (BI) is an important category of software that can help an organization turn raw data into usable information to aid in better decision-making. Microsoft Power BI is one such tool that offers formidable data preparation, data visualization, and interactive dashboard-building options. Power BI transforms data into interactive visuals, allowing companies to uncover trends, make comparisons and share insights more effectively.

This project involves analyzing a dataset of hotel transactions that includes data on hotel bookings, sales, customer segments, booking channels, hotel location data, customer ratings, and booking dates. It was then imported into Microsoft Power BI Desktop to perform cleansing, transformation and analysis of the data for all the relationships between the various business variables and meaningful patterns.

The main goal of this analysis will be to make interactive dashboards to visualize performance in the hotel business and customer behavior. In order to reach this goal, the dataset was investigated with different types of visualizations and ten of the most content-relevant and informative visualizations were chosen to best communicate the main findings. Microsoft Dynamics 365 CRM provides the two dashboards mentioned above: Executive Dashboard, which is business performance-centric and Customer Insights Dashboard, which is customer-centric and offers in-depth insights about bookings, customer segments, and service performance levels.

Visualizations are not presented in isolation, rather this report takes a data storytelling approach that incorporates a series of visualizations to create a connected story. For each visualization chosen, the report teases out the insight that can be gained from the visualization, why these were the most suitable visualization type, and why and what service they contribute to the business story. This fits in with the goal of data transformation whereby data from transactions are translated to actionable business data for informed managerial decision making.

2. Dataset Exploration

To gain an understanding of the overall data quality and to explore the dataset for a comprehensive understanding, Microsoft Power BI was used at the first stage. Several exploratory plots and analyses were produced to look at the distribution of the data itself, relationships between variables and inconsistencies that may impact the analysis before the final visualizations were selected. The exploratory process resulted in the adoption of visualizations for which there were dependable and meaningful data.

In the data preparation phase, the data was taken into Power BI and preprocessed through Power Query. The column names were checked, data types determined and text cleanup (removing trailing or leading spaces) was performed. Data Quality checks also carried out to ensure that data have appropriate value for data analysis. These preparation steps helped to make sure the data set was consistent and to use it correctly during the process of creating the dashboard.

The exploratory analysis revealed a number of key properties of the data from hotel bookings. The sales transactions were spread over three different hotels to enable meaningful comparisons of the performance of business between hotels and geographical area. There was also details about booking channels, customer segments, arrival dates, average daily rate (ADR), customer ratings and length of stay, multiple angles were used to analyze the business.

Several possible visualizations were developed during exploration to understand, among others, the distribution of revenue, customer behavior, booking information and service quality. This included all the exploratory visualizations for the study but not every visualization was included in the final report to include the ten visualizations which best communicated the key business insights. These visualizations all tell a story by providing a view of all business performance, regional comparisons, customer segments, booking trends, seasonal performance and customer satisfaction.

The exploratory stage was used to build a foundation for the design of the dashboard. The exploratory stage was used as a basis and a foundation for the design of the dashboard. The analysis started with a way of understanding the set of data before developing the final report, so that the analysis not only produced descriptive charts but also investigated meaningful questions for business. This way, everyone selected was helping to tell a business story based on evidence and a defined approach.

3. Visualization Selection & Analysis

3.1 Executive Dashboard

The Executive Dashboard was created to give management a quick summary of how the hotel's various areas of business are doing overall. Summarizes the financial performance, operational activity, pricing efficiency, customer satisfaction, regional performance and sales trends in a combination of four Key Performance Indicators (KPIs) and three analytical visualizations. The dashboard provides decision makers with a quick overview of the business situation before diving into in-depth customer-centric analyses.

3.1.1 Total Sales (KPI Card)

Insight

From the Total Sales KPI, the net revenue that the hotels earned in the timeframe of the data set was around RM 46.78 million. The overall financial performance of the hotels and is the most important indicator of business success. It makes it possible to give management an instant view on all payments received for hotel reservations.

Rationale

The KPI Card was chosen because it is specifically designed to present one high-level KPI. They should be presented via a KPI card rather than a bar chart or table, which is recommended for comparisons rather than showing a single value.

Contribution

This visualization gives an overview of the hotels' overall financial outcome, to serve as the initial point of the analysis. This revenue is the basis for the other visualizations, allowing to understand how this revenue is spent by various groups of hotels, regions, customers, booking channels, and time periods.



Figure 1: Total Sales KPI

3.1.2 Total Bookings (KPI Card)

Insight

The Total Bookings KPI indicates that there are about 6,000 bookings in the dataset. The overall demand for hotel services is one of the measures that reflect the overall customer activity in the time interval analyzed. It can also be compared with the total sales to see how the business is performing on volume and the money it can produce.

Rationale

The KPI Card was chosen, since booking volume is among the most important overall indicators, which, when opened, should be immediately visible on the opening screen of the dashboard. A KPI card will help users quickly process the information without having to perform a comparison between different categories or time periods.

Contribution

This visualization along with the Total Sales KPI can help you see the activity level which reflects the volume of business that contributed to your reported top line. Both these KPIs give operators an overview of the performance of their hotels before going into more specific analyses.

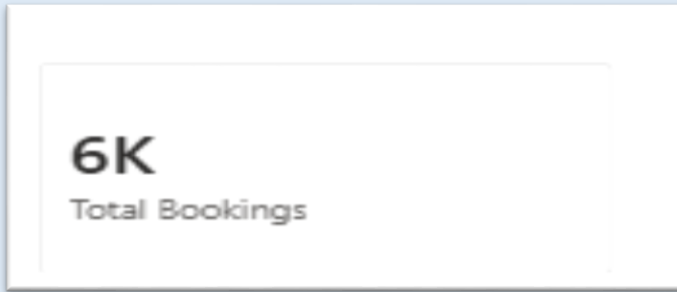


Figure 2: Total Bookings KPI

3.1.3 Average Daily Rate (Average ADR)

Insight

Average Daily Rate (ADR) KPI is an Average of RM 84.59 per occupied room. It's one of the most critical KPIs in the hotel business as it shows how well the hotel is performing on pricing and how much is being made from room sales.

Rationale

A KPI Card was used because ADR is a single measure: numeric value, and there is no need to compare between category. As a KPI it provides the managers with an instant visualization of pricing performance without any unnecessary complexity.

Contribution

This visualization takes financial analysis one extra step further with the addition of pricing efficiency. In addition to Total Sales which represents overall revenue, ADR represents the average revenue generated from each room occupied and serves as an indicator of the performance of a hotel's revenue.

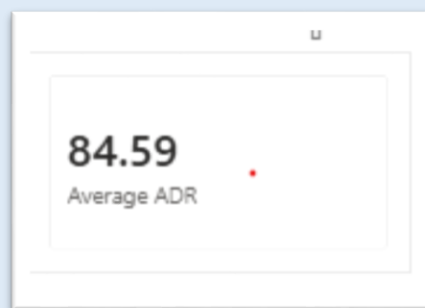


Figure 3: Average Daily Rate KPI

3.1.4 Average Customer Rating

Insight

The Average Customer Rating KPI reports a satisfactory customer rating of about 3.69 on the hotels on average. One of the Key Points to consider for Non-Financial Performance indicator would be the satisfaction of the customers, as it indicates the quality of the service provided by the hotels.

Rationale

Using a KPI Card to select a single summary value was chosen as the best way to show average customer satisfaction. A KPI can be used to get a rapid overview of service quality to be compared before getting into more detailed visualizations of service related to customers.

Contribution

This visualization is an addition to the Financial and Operation KPI and adds customer perspective to the dashboard. All four KPI cards give a well-rounded analysis of the business, as revenue, operational business activity, pricing effectiveness and customer satisfaction combine to create an all-round executive summary.

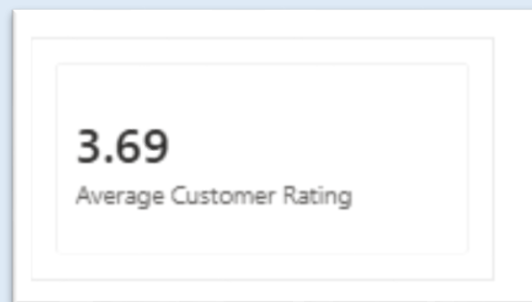


Figure 4: Average Customer Rating KPI

3.1.5 Sales by Hotel



Figure 5: Sales by Hotel

Insight

Sales by Hotel visualization shows all the sales amounts of each hotel that were included in the dataset. The chart allows for a clear of comparison of hotel performance through showing the revenue contribution of each hotel. From the dashboard, sales statistics of the three hotels are quite comparable with a difference in total revenue. The comparison can help the management determine which hotel is performing best, and if there is a hotel that is not finding business opportunities, it can be identified.

Rationale

Use of clustered bar chart was decided since it is one of the most suitable graphical representations for comparing numerical values within different categories. The bar chart can be used to see at a glance and check the numbers to see how the sales of the three hotels compare. The bar chart offers the best visual comparison of the data provided compared to a pie chart which shows proportions, or a table which takes longer to read through and interpret.

Contribution

This visualization takes the overall business performance analysis one step further by taking a look at individual hotel performance. This chart communicates the total revenue information given in the KPI cards and provides management with an understanding of where revenue has been made in the business, to enable them to analyse the contribution of each hotel to the organisation's overall financial performance.

3.1.6 Sales by Region

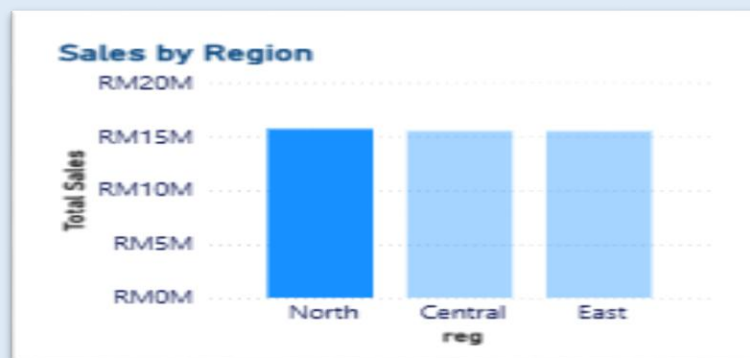


Figure 6: Sales by Region

Insight

The Sales by Region visualization displays the sales for a hotel by region. It shows revenue differences between store locations, and helps management determine which stores are more important to its bottom line. Knowledge of the regional performance helps decision makers to assess market demand and thereby locate potential business areas.

Rationale

A clustered column chart was chosen as it clearly illustrates quantitative data for more than one category and shows the differences between regions. Vertical columns also offer a visual perspective of sales activity by region and enable a user to easily compare categories without needing to know at a granular level how each category is performing.

Contribution

This visualization will add a geographic viewpoint to the hotel performance comparison that was done earlier. It displays areas of the business's strength, which allows for location-based decision making, resource allocation, and future expansion plans.

3.1.7 Monthly Sales Trend

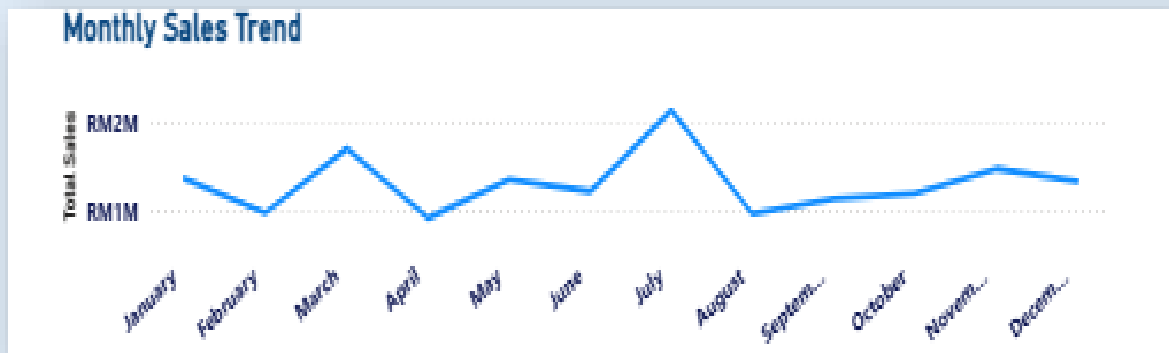


Figure 7: Monthly Sales Trend

Insight

Monthly Sales Trend visualization shows the overall sales trend for the entire reporting period. The line chart is useful for tracking month-to-month income to see the ups and downs of sales volume, helping management identify peaks and troughs in sales. This trend can be attributed to seasonal demand, promotional efforts, customer dynamics and travel habits, shopping behavior, or various other business factors that affect hotel performance.

Rationale

The chart type chosen was a line chart, as this is the best chart to illustrate change over time. A line chart highlights the continuity of data, making it easier to see increases or decrease in sales. A line chart is used in the same capacity as a bar or pie chart for comparing categorical data. This enables the user to better see the patterns and trends.

Contribution

This visualization is used to enhance the Executive Dashboard further by adding the time dimension to the analysis. The monthly trend clarifies when the revenue changed, previous explanations

informed on where and how much of revenue was generated. Combining the KPI cards, the hotel comparison and the regional comparison, and the monthly trend gives a holistic view into overall business performance.

3.2 Customer Insights Dashboard

The Customer Insights Dashboard accompanies the Executive Dashboard to highlight the behaviour and bookings of the customers instead of the overall financial results. The Executive Dashboard provides answers to questions on revenue, bookings, pricing and regional performance, whilst the Customer Insights Dashboard focuses on aspects that affect these results. The visualizations identified are the results of customer segmentation, booking type analysis, and customer satisfaction analysis to facilitate service management to understand the customer's preference and satisfaction level. Such experienced knowledge can be used to enhance marketing strategies, customer relationships, and operations.

3.2.1 Sales by Customer Segment

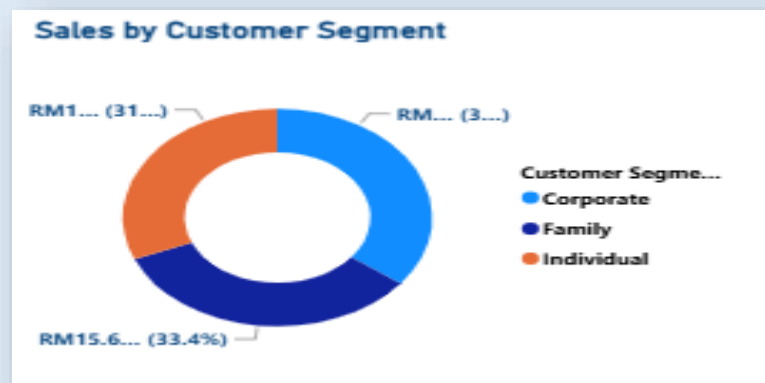


Figure 8: Sales by Customer Segment

Insight

The Sales by Customer Segment visualization displays the amount of sales for each customer segment. This comparison allows the management to find out which is the most important group to help the hotel build its revenue. Knowing what types of customers there are is important because their booking habits, spending patterns and expectations are likely to vary. The image therefore can offer useful information to analyze financial contribution of each customer segment.

Rationale

A donut chart was chosen as the visualization tool as the goal of the visualization is to show what percentage of total sales comes from each group of customers. Donut charts, as opposed to bar charts, highlight the value of each segment in relation to the total revenue; they make it easy to grasp the pie chart's meaning. There are a relatively small number of customer segments to represent, making the donut chart an effective and intuitive way to represent these segments.

Contribution

This visualization takes the focus of analysis from organizational performance to customer behavior. Once overall revenue and business performance was identified on the Executive Dashboard, it provides insights into which customer segment(s) is generating revenue, allowing for greater insights to be provided around customer relation management, market segmentation and targeted marketing strategies.

3.2.2 Sales by Booking Channel



Figure 9: Sales by Booking Channel

Insight

The Sales by Booking Channel visualization takes a tactic from the Booking Channel context menu and presents the revenue from various booking channels. The chart illustrates the extent to which each booking source contributes and illustrates how customers book their hotel. Knowing the performance of each booking channel helps manage the efficiency of various distribution channels and take strategic measures to enhance marketing strategy or the partnership.

Rationale

The type of chart chosen was clustered bar chart as it is used to compare sales values from each booking channel. When labels are variable length, horizontal bars can be used to create clear, easily-understood charts that can be used to compare the difference in revenue. A bar graph offers a more accurate representation of the performance of the booking channels compared to pie charts or tables.

Contribution

This visualization gives an explanation of how the customer can interact with the hotel services. The last visualization showed you who is responsible for revenue but this visualization is showing you the methods used to book revenue. These data pieces taken together offer a much better view to

management of their customers' purchasing habits and inform decisions about marketing investment and channel management.

3.2.3 Average Customer Rating by Hotel



Figure 10: Average Customer Rating by Hotel

Insight

Average Customer Rating by Hotel displays the satisfaction level of the guests to the various hotels in the dataset. This feature indicates an average score at each hotel to help management compare services and determine if there are discrepancies in the customer perception of service between hotels. Customer ratings are a good indicator of service quality as they contain information on the entire visit to the hotel.

Rationale

A clustered column chart was chosen due to the fact that it would be effective for comparing the numerical values of data from a limited number of categories. The chart is easy to read and maintains a professional look whilst providing quick and easy identification of differences in customer satisfaction. The column chart gives a better comparison between individual hotels than a table or a KPI card.

Contribution

This visualization completes the Customer Insights Dashboard, adding a "quality" view to the analysis. Customer segments and booking behavior were analyzed and then an assessment of customer satisfaction with services offered by each hotel. The three visualizations make up the whole business story in terms of future customers, their preferences to book and the quality of your services.

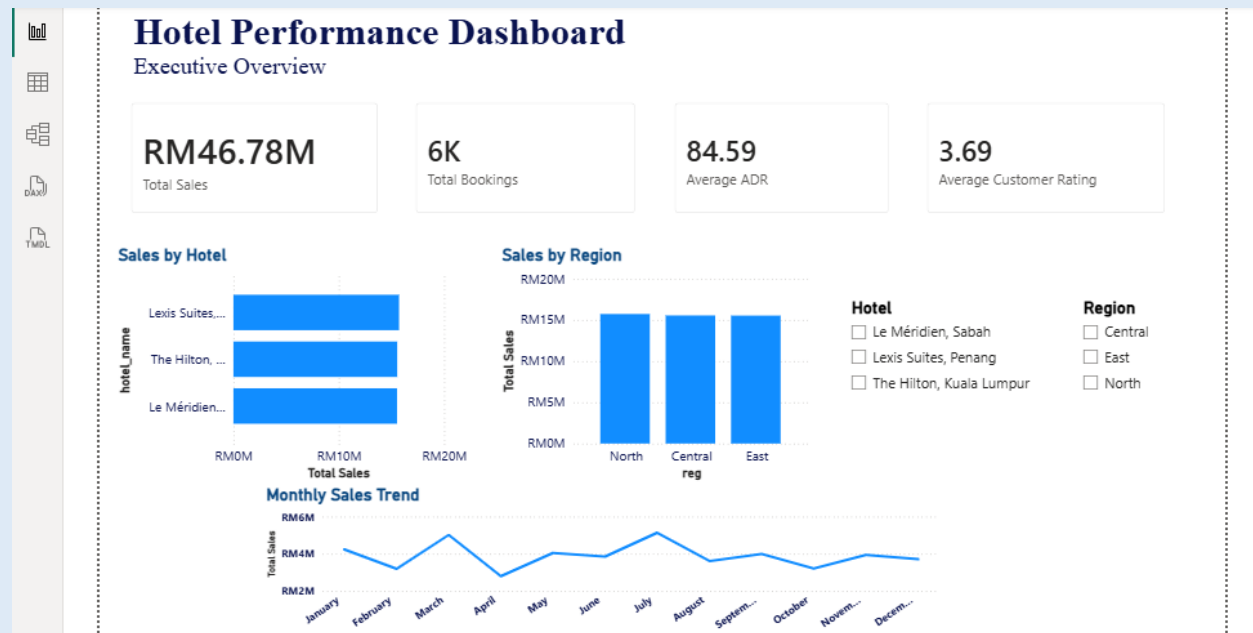
4. Conclusion

The task of this project was to use MS Power BI to provide business insights from a dataset of a transactional hotel. Some important patterns were investigated in the data in regard to hotels' performance result, pattern of customer behavior, and coming pattern of the customer selling from, and the pattern of service quality in the respective hotels. Ten visualisations were selected because it was decided that this number would provide the best representation of the key findings as would help bring the business story to the fore.

The executive Dashboard provided an overview of business performance in a series of KPIs, and provided comparative and trend-based charting. Thanks to these visualizations, the management could view the total revenue, booking volume, pricing and customer satisfaction as well as regional sales distribution and monthly revenue trends. These insights were enhanced by the inclusion of the Customer Insights Dashboard, which took the conversation deeper into customer shop segment, booking channel and hotel ratings but provided additional insights into the customers' behavior and hotel rating.

The featured visualizations showed how interactive dashboards can enable anyone to convert large amounts of transactional information into information that aids managerial decision-making. The dashboards weren't just going to show charts, but were designed to give greater clarity on the business overall and to translate from the "big picture" to the individual customer level. This helps decision makers with the analysis of the performance trends, assess operational effectiveness and make data-driven and informed decisions with regard to the strategic decisions.

5. Appendices

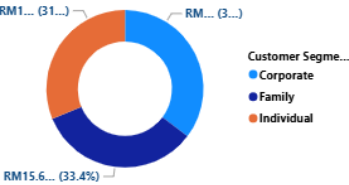




Customer Insights Dashboard

Customer & Booking Analysis

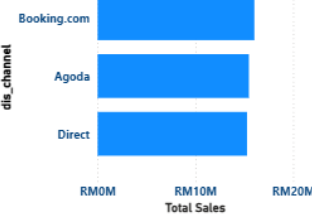
Sales by Customer Segment



Customer Segment

- ☐ Corporate
- ☐ Family
- ☐ Individual

Sales by Booking Channel



Booking Channel

- ☐ Agoda
- ☐ Booking.com
- ☐ Direct

Average Customer Rating by Hotel

